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Physics

Motion in a Plane

DPP: 4

Q1 From a certain height, two bodies are projected horizontally with velocities $10 \mathrm{~m/s}$ and $20 \mathrm{~m/s}$. They hit the ground in t_1 and t_2 seconds. Then

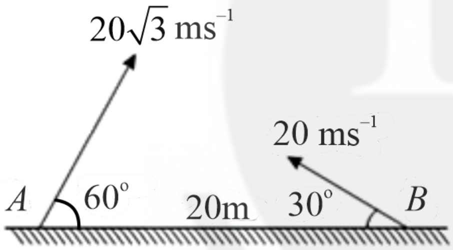
(A) $t_1 = t_2$

(B) $t_1 = 2t_2$

(C) $t_2 = 2t_1$

(D) $t_1 = \sqrt{2t_2}$

Q2 In the figure shown, the two projectiles are fired simultaneously. Find the minimum distance between them during their flight.



(A) 10 m
(B) 20 m
(C) 30 m
(D) None of these

Q3 The position co-ordinates of a projectile thrown from ground are given by $y = (12t - 5t^2)$ m and $x = 5t$ (m). The initial speed of throw is (Here t is in second)

(A) 10 m/s

(B) 13 m/s

(C) 11 m/s

(D) 17 m/s

Q4 A particle is projected at an angle of projection α and after t seconds it appears to have an angle of β with the horizontal. The initial velocity is:

(A) $\frac{gt}{2\sin(\alpha-\beta)}$

(B) $\frac{gt\cos\beta}{\sin(\alpha-\beta)}$

(C) $\frac{\sin(\alpha-\beta)}{2\,gt}$

(D) $\frac{2\sin(\alpha-\beta)}{gt\cos\beta}$

Q5 The speed of boat is $18 \mathrm{~km/h}$ in still water. It crosses a river of width $2 \mathrm{~km}$ along the shortest path in 7 minutes. The velocity of the river is:

(A) $5.5 \mathrm{~km/h}$

(B) $6 \mathrm{~km/h}$

(C) $7 \mathrm{~km/h}$

(D) $10 \mathrm{~km/h}$

Q6 The equation of a projectile is $y = \sqrt{3}x - \frac{gx^2}{2}$. The angle of projection is-

(A) 30°

(B) 60°

(C) 45°

(D) None

Q7 A projectile is launched at an angle 'α' with the horizontal with a velocity 20 m/s. After 10s, it's inclination with horizontal is 'β'. The value of $\tan\beta$ will be: $\left(g = 10ms^{-2}\right)$.

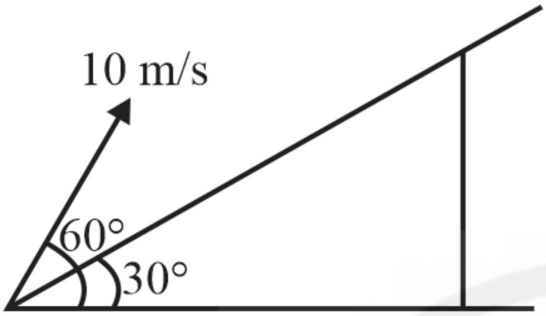
(A) $\tan\alpha + 5\sec\alpha$

(B) $\tan\alpha - 5\sec\alpha$

(C) $2\tan\alpha - 5\sec\alpha$

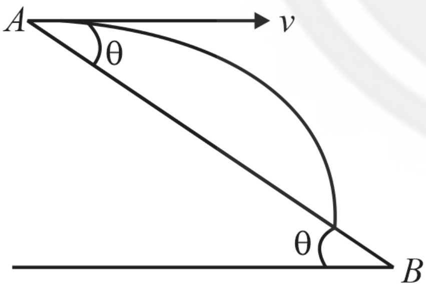
(D) $2\tan\alpha + 5\sec\alpha$

Q8 A particle is thrown at time $t = 0$ with a velocity of 10 m/s at an angle 60° with the horizontal from a point on a inclined plane, making an angle of 30° with the horizontal. The time when the velocity of the projectile becomes parallel to the incline is (g = 10 m/s^2)



(A) $\frac{2}{\sqrt{3}} s$
(B) $\frac{1}{\sqrt{3}} s$
(C) $\sqrt{3} s$
(D) $\frac{1}{2\sqrt{3}} s$

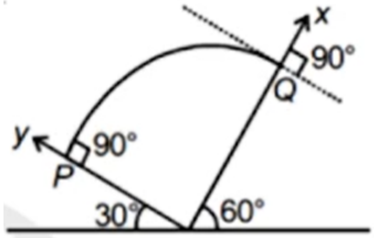
Q9 A particle is projected horizontally with a speed $v = 5 ms^{-1}$ from the top of a plane inclined at an angle $\theta = 37^\circ$ to the horizontal as shown in figure Find the time taken by the particle to hit the plane. {Take g = 10 m/s^2}



(A) $3/4s$
(B) $3s$
(C) $4s$
(D) $4/3s$

Q10 Two incline plane of angle 30° and 60° are placed touching each other at the base as shown in the figure. A projectile is projected with speed of $10\sqrt{3}\ \mathrm{ms}^{-1}$ from point P as shown and hits the other inclined at point Q normally.

If the co-ordinate are taken along the inclines as shown in the figure, then The time of flight in the above problem is



(A) $1\ \mathrm{s}$
(B) $2\ \mathrm{s}$
(C) $3\ \mathrm{s}$
(D) $4\ \mathrm{s}$

Q11 A man standing on an inclined plain observes rain is falling vertically. When he starts moving down the inclined plain with velocity $v = 6\ \mathrm{m/s}$ observes rain hitting him horizontally.

The actual velocity of rain is
(A) $3\ \mathrm{m/s}$
(B) $3\sqrt{3}\ \mathrm{m/s}$
(C) $4\ \mathrm{m/s}$
(D) none of these

Q12 The trajectory of a projectile near the surface of the earth is given as $y = 2x - 9x^2$. If it were launched at an angle θ_0 with speed v_0 then $\left(g = 10\ \mathrm{m\ s^{-2}}\right)$

(A) $\theta_0 = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_0 = \frac{5}{3}\ \mathrm{m\ s^{-1}}$
(B) $\theta_0 = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_0 = \frac{5}{3}\ \mathrm{m\ s^{-1}}$

(C) $\theta_0 = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_0 = \frac{3}{5}\ \mathrm{m\ s^{-1}}$

(D) $\theta_0 = \sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_0 = \frac{3}{5}\ \mathrm{m\ s^{-1}}$

Q13 A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution. If this particle were projected with the same speed at an angle θ to the horizontal, the maximum height attained by it equals $4\mathrm{R}$. The angle of projection, θ, is then given by:

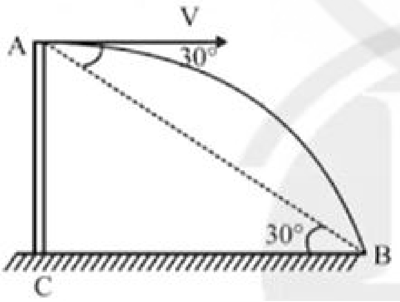
(A) $\theta = \cos^{-1}\left(\frac{\pi^2 \mathrm{R}}{\mathrm{g T^2}}\right)^{1/2}$

(B) $\theta = \sin^{-1}\left(\frac{\pi^2 \mathrm{R}}{\mathrm{g T}}\right)^{1/2}$

(C) $\theta = \sin^{-1}\left(\frac{2g\,\mathrm{T^2}}{\pi^2 \mathrm{R}}\right)^{1/2}$

(D) $\theta = \cos^{-1}\left(\frac{\mathrm{g T^2}}{\pi^2 \mathrm{R}}\right)^{1/2}$

Q14 An object is thrown horizontally from a point A from a tower and hits the ground $3\ \mathrm{s}$ later at B. The line from A to B makes an angle of 30° with the horizontal. The initial velocity of the object is : (take $g = 10\ \mathrm{m/s^2}$)



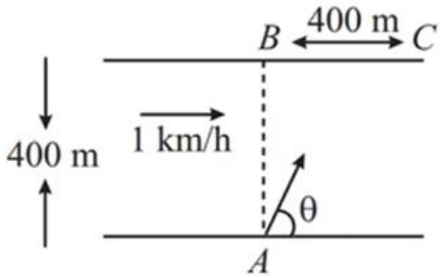
(A) $15\sqrt{3}\ \mathrm{m/s}$

(B) $15\ \mathrm{m/s}$

(C) $10\sqrt{3}\ \mathrm{m/s}$

(D) $25/\sqrt{3}\ \mathrm{m/s}$

Q15 A river is flowing with a speed of $1\ \mathrm{km/hr}$. A swimmer wants to go to point C starting from A. He swims with a speed of $5\ \mathrm{km/hr}$, at an angle θ w.r.t. the river flow. If $AB = BC = 400\ \mathrm{m}$. At what angle (θ) with river bank should swimmer swim?



(A) 30°

(B) 37°

(C) 53°

(D) 60°

Q16 Two objects are projected horizontally in opposite directions from the top of a tower with velocities u_1 and u_2. The time when their velocity vector are perpendicular to each other is

(A) $\frac{u_1 u_2}{g}$

(B) $\sqrt{\frac{u_1 u_2}{g^2}}$

(C) $\frac{u_1 u_2}{2g}$

(D) $\sqrt{\frac{u_1 u_2}{2g^2}}$

Q17 A plane surface is inclined making an angle θ with the horizontal. From the bottom of this inclined plane, a bullet is fired with velocity v. The maximum possible range of the bullet on the inclined plane is

(A) $\frac{v^2}{g}$

(B) $\frac{v^2}{g(1+\sin\theta)}$

(C) $\frac{v^2}{g(1-\sin\theta)}$

(D) $\frac{v^2}{g(1+\cos\theta)}$

Q18 When it is raining vertically down, to a man walking on road the velocity of rain appears to be 1.5 times his velocity. To protect himself from rain he should hold the umbrella at an angle θ to vertical. Then $\tan\theta =$

(A) $\frac{2}{\sqrt{5}}$

(B) $\frac{\sqrt{5}}{2}$

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

Q19 A projectile is thrown with an initial velocity of $\vec{v} = a\hat{i} + b\hat{j}$. If range of the projectile is double the maximum height attained by it then:

(A) $a = 2b$

(B) $b = a$

(C) $b = 2a$

(D) $b = 4a$

Q20 A projectile is fired at angle θ with horizontal. When the particle makes an angle β with the horizontal, its speed becomes v. v is given by

(A) $v = u\cos\theta \cdot \sec\beta$

(B) $v = u\cos\theta \cdot \sin\beta$

(C) $v = u\cos\theta \cdot \cos\beta$

(D) $v = u\cos\theta \cdot \operatorname{cosec}\beta$

Q21 Average velocity of a particle in projectile motion between its starting point and the highest point of its trajectory is: (projection speed $= u$, angle of projection from horizontal $= \theta$)

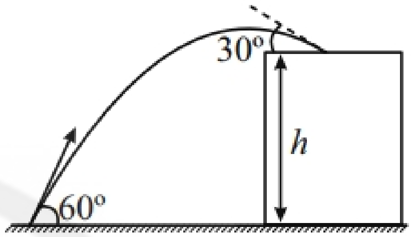
(A) $u\cos\theta$

(B) $\frac{u}{2}\sqrt{1+3\cos^2\theta}$

(C) $\frac{u}{2}\sqrt{2+\cos^2\theta}$

(D) $\frac{u}{2}\sqrt{1+\cos^2\theta}$

Q22 A stone projected at an angle of 60° from the ground level strikes at an angle of 30° on the roof of a building of height ' h '. Then the speed of projection of the stone is:



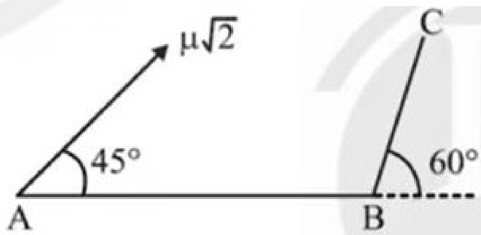
(A) $\sqrt{2gh}$

(B) $\sqrt{6gh}$

(C) $\sqrt{3gh}$

(D) $\sqrt{gh}$

Q23 A particle is projected from a point A with velocity $u\sqrt{2}$ at an angle of 45^{0} with horizontal as shown in figure. It strikes the plane BC at right angles. The velocity of the particle at the time of collision is



(A) $\frac{\sqrt{3}u}{2}$

(B) $\frac{u}{2}$.

(C) $\frac{2u}{\sqrt{3}}$

(D) u

Q24 Match the following:

	Column I		Column II
(A)	Time for a boat to cross a river of width (by the shortest distance ($\vec{v}$ -velocity of boat with respect to water $\vec{u}$ - velocity of water)	(P)	$\frac{\ell}{\lvert\vec{v}+\vec{u}\rvert}$
(B)	Time for two particles moving with velocities $\vec{v}$ and $\vec{u}$ in opposite direction to meet each other. (initial separation of particles is C)	(Q)	$\frac{\ell}{\sqrt{v^2-u^2}}$
(C)	Time for a boat to cross a river of width C in the shortest time ($\vec{v}$ - velocity of boat with respect to water, $\vec{u}$ - velocity of water)	(R)	$\frac{\ell}{\lvert\vec{v}\rvert+\lvert\vec{u}\rvert}$
(D)	Time for a boat to travel a distance ℓ downstream ($\vec{v}$ - velocity of boat with respect to water, $\vec{u}$ - velocity of water)	(S)	$\frac{\ell}{\lvert\vec{v}\rvert}$
		(T)	$\frac{\ell}{\sqrt{u^2+v^2}}$

(A) $A-P \quad B-Q, R \quad C-T \quad D-S$

(B) $A-P, Q \quad B-R \quad C-S \quad D-P, R$

(C) $A-Q \quad B-P \quad C-R \quad D-T$

(D) $A-Q \quad B-S \quad C-T \quad D-P, S$

Q25 The path of projectile is represented by

$y=Px-Qx^2$.

	Column- I		Column-II
i.	Range	a.	P/Q
ii.	Maximum height	b.	P
iii.	Time of flight	c.	$P^2/4Q$
iv.	Tangent of angle of projection is	d.	$\sqrt{\frac{2}{Qg}}P$

(A) $i-a\ ,\ ii-c\ ,\ iii-d\ ,\ iv-b$

(B) $i-c\ ,\ ii-a\ ,\ iii-d\ ,\ iv-b$

(C) $i-a\ ,\ ii-d\ ,\ iii-c\ ,\ iv-b$

(D) $i-a\ ,\ ii-d\ ,\ iii-c\ ,\ iv-b$

Answer Key

Q1	A
Q2	A
Q3	B
Q4	B
Q5	A
Q6	B
Q7	B
Q8	B
Q9	A
Q10	B
Q11	B
Q12	A
Q13	C
Q14	A
Q15	C
Q16	B
Q17	B
Q18	A
Q19	C
Q20	A
Q21	B
Q22	C
Q23	C
Q24	B
Q25	A